Experiment 7

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Subject Name: System Design
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1. **Aim**: To design and develop a ride-sharing system that enables users to book rides and drivers to accept requests with real-time tracking and efficient matching.

2. **Objective**:

- Provide on-demand ride booking
- Enable real-time driver allocation
- Ensure fast and reliable matching
- Support secure payment system
- Provide live tracking & navigation

3. **System Requirements:**

A. Functional Requirements

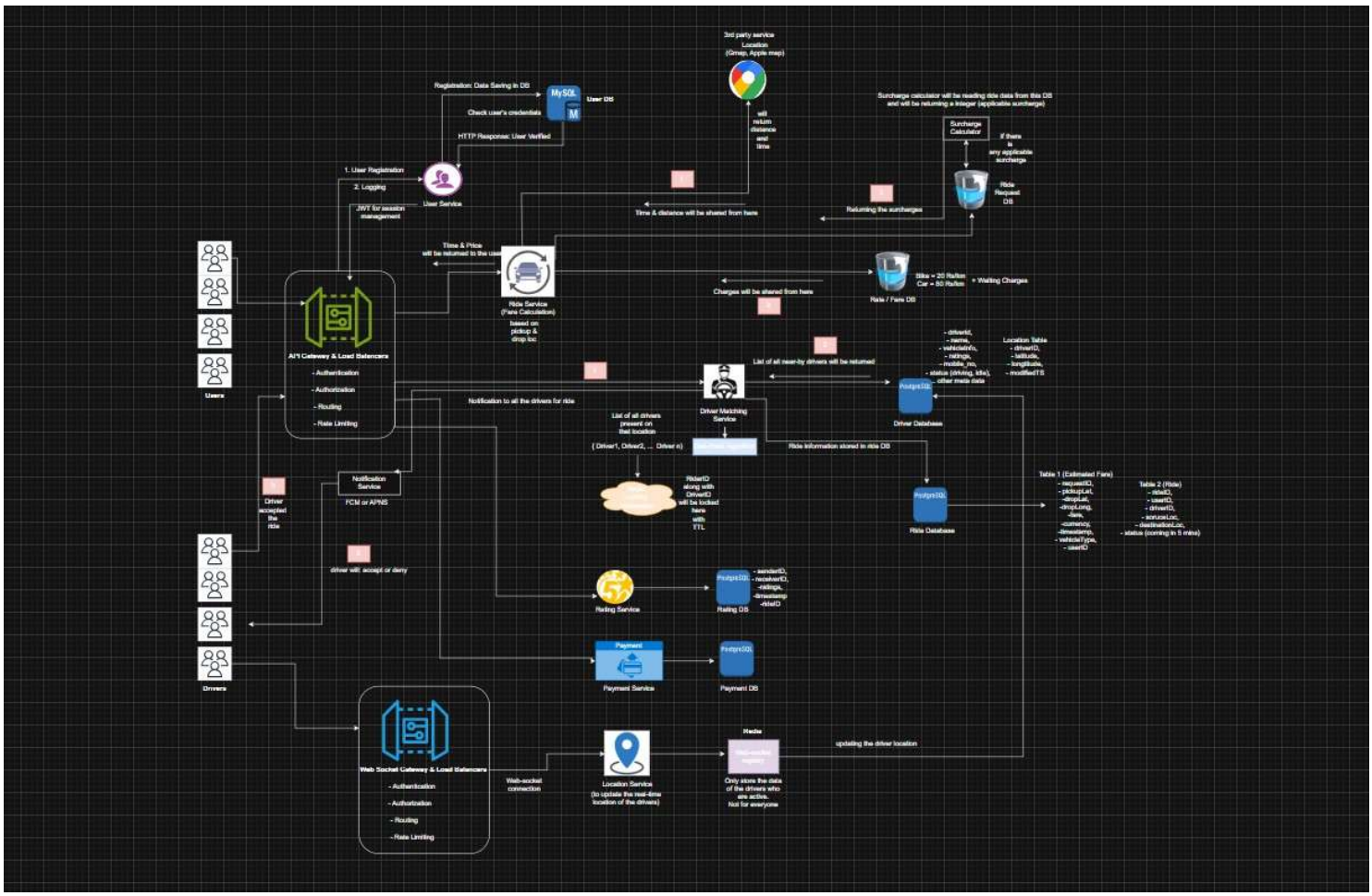
- **User (Rider)**
  a. Get fare estimation
  b. Request ride
  c. Choose ride type (car, bike, etc.)
  d. Track ride in real-time
  e. Make payments
  f. Rate driver
- **Driver**
  a. Accept / reject ride
  b. Navigate to pickup & drop
  c. Update location continuously
  d. Complete ride
- **Core System**
  a. Match rider with nearby driver
  b. Real-time tracking
  c. Ride history
  d. Notifications to drivers
  e. Users should be able to search videos based on title, creator name, or tags.

B. Non-Functional Requirements

- **Scalability**
  Target: ~1.5 million users & drivers
- **Performance**
  Driver assignment within < 1 minute
- **CAP Theorem**
  Strong consistency → driver not assigned to multiple users
  Eventual consistency → for user data / updates Availability
  High availability for users (system always accessible)

- **Latency:**
  Video playback latency should be **low (around 50–80 ms)** with minimal buffering during streaming

4. Low Level Design (LLD):



5. Scalability Solution

- Use Microservices Architecture (separate services for user, driver, ride, payment)
- Implement Load Balancers to distribute traffic
- Apply Database Sharding (split data by region/location)
- Use Database Replication for high availability
- Implement Caching (Redis) for faster access to frequent data
- Use Geo-Spatial Indexing for efficient nearby driver search.

6. Learning Outcomes (What I Have Learnt)

- Understand system design of real-world apps (Ola/Uber)
- Learn functional vs non-functional requirements
- Gain knowledge of scalable architecture design
- Understand CAP theorem (Consistency vs Availability)
- Learn real-time data handling (location tracking).